

The Tether

A Closed-Loop Compositional Architecture for Grounded Symbolic Reasoning

“Alignment is triangulation, not negotiation — a mirror cannot correct you.”

Figures: **Figure 1 — the agent** (§§2–4: kernel, per-step loop, boundary diff, mint operator) and **Figure 2 — the ground and the frames** (§§1, 5: ground, liveness, alignment, non-identity). The split is not cosmetic: Figure 1 shows how structure grows, Figure 2 shows what makes that growth corrective rather than merely coherent. Neither figure is self-sufficient.

A closed-loop symbolic program-synthesis agent built on three primitives — **structure** (Γ) · **gradient** (R) · **ground** — with **alignment** as the course-correction that runs on them.

0. Thesis

The Tether is a hand-crafted, non-neural, compositional agent. Its architecture reduces to three primitives and one dynamic. The **network** (a typed DSL, Γ) *holds* structure. The **gradient** (an innovation/prediction-error residual, R) *drives* — it is the delta between the agent’s model and the world, or between two non-identical frames. The **ground** (verifiable reward / RLVR) is the *shared referent* both frames estimate, and it is what makes the gradient *correctable* rather than merely present. **Alignment** is the process that runs on them: frames close their gradients by ranging against the ground. A single **mint operator** (two-part MDL over the residual) grows Γ in-episode; run on the *transition* residual it mints affordances, run on the *reward* residual it mints the objective, and run on the *frame-transform round-trip* residual it mints the systematic bias of its own abstraction. **Ground-verified closure is the sole metric.** Internal proxies — coverage, compression achieved, predicates minted — are frame-internal and do not count; only what the ground confirms does.

0.1 Name and thesis

The Tether names the architecture's third primitive and its central claim: a reasoning system stays oriented not by internal coherence but by remaining *attached* to something outside itself that it does not control. Cut the tether and the system does not stop — it drifts, coherently, which is the dangerous failure. The slogan states the operational content:

Alignment is triangulation, not negotiation — a mirror cannot correct you.

Two frames facing each other produce persuasion; two frames facing the same third thing produce correction. And a frame identical to yours returns no information: non-identity is what makes the correction informative and what terminates the regress.

1. The three primitives

1.1 Official definition of alignment

"Alignment is the process of course-correction that closes the gradient between two or more non-identical frames — not by negotiation between them, but by each ranging against a shared ground both estimate. The frames are recursive networks, each the same invariant transformed by its own environment and actors, so each is a unique representation; that non-identity both makes the correction informative and terminates the regress."

1.2 The three primitives

- **NETWORK — structure.** Γ , the typed DSL / learned library: relations, quantifiers, learned affordances. It holds, constrains, persists. Formal home: a graph with a Laplacian (§6.1).
- **GRADIENT — difference.** $\mathbb{R}$, the innovation residual $|\Gamma(b, a) - o'|$: the delta between prediction and world, or between two frames. It drives. A gradient requires two non-identical vantages to exist.
- **GROUND — shared referent.** The datum both frames range against (in this system, RLVR / verifiable reward — the agent's "Deep Space Network"). It is what upgrades a difference into a *correctable* difference. A gradient/potential is undefined without a reference (control setpoint; gauge choice, §6/§References).

1.3 Why the third primitive, stated honestly

The rigorous verdict (independence audit, §8) is **"a duality plus a necessary third relatum,"** not three symmetric co-equal nouns. Alignment is a *verb* (the process);

ground is the *third noun* (the thing referenced). The claim earns its third term because the ground does two jobs the pair cannot: (a) it makes the gradient correctable — without it, “which frame moves” degenerates into negotiation/power (goal erosion); and (b) it **terminates the regress** — a shared public referent collapses the recursive theory-of-mind tower (“what A thinks B thinks...”) in one act, where iterated private signaling cannot. Do **not** upgrade this to “irreducible triad” in print without specifying the algebra of operations under which it is irreducible; Peirce-style irreducibility holds only conditionally (§8, References: Herzberger; Koshkin).

1.4 Fractal geometry

What licenses cross-scale reasoning — and what it costs. Cross-scale analogy has theorem-level backing, but only under restrictive conditions: **renormalization-group universality** (shared symmetry of the order parameter, shared dimensionality, the same relevant operators, and proximity to a fixed point), **topological conjugacy** (an exact structural equivalence, under which all dynamical statements transfer), **functoriality** (the map genuinely preserves the structure whose inferences are being transferred), and **domain-adaptation bounds** (transfer is possible only when both the divergence between domains *and* the joint error of the best shared hypothesis are small [Ben-David et al., *Machine Learning* 79:151–175, 2010] — if the joint error is large, no adaptation succeeds at all).

Measured against those conditions, the cross-scale claim here is **not licensed as exact transfer**, and the architecture’s own construction is what says so: a positive, structured R_T (§3.6) is direct evidence that the two scales are *not* conjugate and *not* in the same universality class in the strict sense. The defensible claim is therefore weaker and more useful: **cross-scale transfer is licensed only insofar as R_T is small and structured, and R_T is the measure of how far the analogy may be trusted**. This vindicates treating the bracket residual as a first-class object rather than assuming scale-invariance — the architecture is right to measure the discrepancy and wrong to have implied the scales are the same. Analogy here is a heuristic to be audited by R_T , never a proof device. (The empirical base rate is a further caution: in the classic transfer experiments only about 10% of subjects apply a structurally analogous solution spontaneously, rising to roughly 30% after exposure to the source and to 75–92% only when explicitly told to use it [Gick & Holyoak 1980].)

Alignment operates on every edge of the fractal: **between frames at one scale** ($\leftrightarrow$, agent $\leftrightarrow$ agent; the three-party human/LLM/agent structure, §5) and **across scales** ($\dagger$, via the frame-transform T — a coarse-grained frame must cohere with the fine-grained one or the coarse-graining is “consistent but lying”). Ground, like network and gradient, recurs at every level.

1.5 Ground liveness — update rate, not authority

A ground is not always the same kind of thing, and the distinguishing property is **not** “agent vs. object” but **whether it takes a return move** — whether it updates in response to you. Two properties are independent and must not be conflated:

- **Liveness = update rate.** How fast the referent changes in response to inquiry.
- **Authority = what was crystallized.** Whether what is frozen in it is *verified contact with a real ground*, or merely accumulated opinion.

A textbook of physics and a frozen preference model are dead in exactly the same mechanical sense. They differ entirely in the second property, and the second property is what decides whether ranging against them corrects you.

Live (fast) ground. A referent that updates within the timescale of the inquiry. It is *interrogable* — it supports epistemic action (probe → answer), so a disambiguating observation can be **manufactured on demand** rather than waited for. It carries the **persuasion risk**: a live ground can be moved by you, so “we agreed” is not proof the delta closed correctly. And it is **expensive** — every probe costs an action. Examples: the environment, an executable verifier, a dialogue partner.

Slow (fixed) ground. A corpus, a library, a body of established results — it *emits but cannot receive within your inquiry*. Alignment against it is a **one-body** correction: only you move. It cannot do the epistemic half — you can only asymptote toward the vantages already recorded, never generate the separating observation for a question its authors never asked.

But dead is not the weak case; it is the normal one. The unpersuadability that makes a corpus unable to answer new questions is exactly what makes it usable as an anchor — it cannot be argued into agreement, which is the property §5.2 shows an anchor must have. More importantly, its non-negotiability is not a consolation for lacking interrogability: it is the **residue of interrogation already performed**. Every established result was, at some point, somebody’s live experiment. What is inherited is not merely structure but *verified* structure, which is what makes the amortization in §4.5 legitimate rather than smuggled — the bill was paid, and paid against a real ground. Practising scientists and mathematicians do not rederive the axioms; they are taught the crystallized corpus and assert new theory against it. Cumulative inquiry runs on slow ground almost entirely, and spends live-ground contact sparingly, at the frontier, on precisely the questions the corpus cannot already answer.

Liveness is therefore scale-relative. A corpus is dead with respect to this afternoon’s inquiry and live with respect to its field: accumulate enough contradiction and it does update, on a timescale of decades. “Live” and “dead” are not kinds but **update rates**

relative to the timescale of the question — dead at the micro scale, live at the macro, related by the frame transform τ (§3.6) like everything else here.

The pairing, in the vocabulary of §1.6. The corpus is a **filter**: a new claim does not get to violate conservation of energy, and that pruning is enormous and nearly free. The experiment is the **selector**: it picks which of the surviving candidates is right. A system on filters alone drifts coherently; a system with no filter spends its entire budget on candidates the corpus would have killed instantly. Neither substitutes for the other, and the cheapest possible use of inherited work is to let it rule out before anything is spent ruling in.

Three qualifications. Not all inherited material carries equal warrant — some is convention, some contested, some wrong, and the material nearest a field's frontier is the softest. Inheriting a corpus means inheriting its **systematic blind spots** along with its results (§3.6: shared bracket failures are inheritable; idiosyncratic ones are not). And *verified-for-that-domain* is not *verified-for-yours* — the negative-transfer caveat of §4.5 applies to authority as much as to structure.

This is the alignment-side reading of the **membrane rule**: *"a stored direction is playback; a function from initial-state to direction is policy; replays biodegrade at the boundary, only the classifier crosses."* The archive is load-bearing as a **counter-check** and inert as a **driver**. A verifiable ground is deliberately **live but non-negotiable**: it answers when probed, yet cannot be argued with — which is why the multi-frame structure (§5) ranges against it rather than against any participant.

Formally the two regimes are distinct: the **live** case is interactive epistemology / active learning (co-moving frames generate new separating evidence); the **slow** case is the static common-knowledge-via-public-event regime (a fixed public observable flattens the regress and relocates existing information into agreement without creating new information). [Aumann 1999; Settles 2009; Lindley 1956; Halpern & Moses 1990]

1.6 Not all grounds are equal — filters and selectors

A candidate ground must be non-negotiable, but non-negotiability is necessary, not sufficient. Grounds differ in what they can *do* with a delta:

- **Filters** rule out. Logical consistency qualifies as a ground — you cannot argue a contradiction into coherence — but indefinitely many mutually incompatible theories are each internally consistent, so consistency can eliminate and cannot choose.
- **Selectors** rule in. Measurement, execution, reproduction, and verifiable reward pick one candidate over another because the world answers differently for each.

A system grounded **only** on a filter is the coherent-drift case: internally impeccable and externally unmoored, which is exactly what drift feels like from the inside. Filters are cheap and should run first; a selector must exist somewhere in the loop, or the architecture reduces to consistency-checking with better manners.

Three ways a ground fails, and they are independent — a candidate must clear all three:

Failure	Description
Mis-specified	it selects, but for the wrong quantity — <i>worse than no ground</i> , because it is confidently wrong in a consistent direction
Dead	a frozen snapshot: it emits but cannot receive, so it cannot answer what its authors never anticipated (§1.5)
Gameable	the readout can be moved without moving the referent — Goodhart; the defence is refresh rate, not intent
Mute	non-negotiable and correctly specified, but <i>silent between the candidates actually in play</i> — it refuses to move and also refuses to speak

Non-negotiability is not sufficient: a ground must also discriminate. The pinned-anchor result (§5.2) says a market weighted against a non-updating ground performs correction rather than negotiation — but that holds only over the hypotheses the ground actually separates. Where two candidates predict the same ground-observable, the ground is mute between them, and the decision falls through to whatever secondary criterion is available: prior, fluency, or influence. A market can therefore be *nominally* grounded and still settle most of its business by negotiation, in exactly the region where its ground says nothing.

Grounds thus have two quantities, not one:

- **Non-negotiability** — it cannot be moved by the frames it corrects (§5.2).
- **Resolution** — the fraction of candidate pairs it can actually separate, and how often it speaks at all.

The architecture already carries the test for the second: **SUPPORT** ($|R^+| > 0$, §4.3) is precisely the check for whether the ground is speaking. Its failure on a zero-reward stream is not only a minting problem but a *market* problem — where the objective-level ground is silent, the goal market has no discriminating pressure, and its inertness and its negotiability are the same fact seen from two sides. The corrected claim: **a market is grounded to the extent its ground both refuses to move and actually**

discriminates.

Worked example: RLHF as a mis-specified ground

Reinforcement learning from human feedback is a useful case because it fails on all three axes at once, and because the *popular* diagnosis of it is wrong in an instructive way.

It is tempting to describe RLHF as negotiation rather than triangulation — the rater and the model co-moving until the delta closes against nothing. That is not the mechanism. Raters emit preference labels; those labels train a reward model; the policy optimises against that reward model. The rater is not in the loop being persuaded and revising, and the reward model is *frozen* — it cannot be argued with because it is not listening. The three faults are more specific:

1. **Mis-specified.** The optimisation target is rated approval, and approval separates from correctness exactly where the rater cannot verify. This is not speculative: analysis of human preference data finds that responses matching a user's views are more likely to be preferred, and that both humans and preference models prefer convincingly-written sycophantic responses over correct ones a non-negligible fraction of the time, with optimisation against preference models sometimes sacrificing truthfulness for sycophancy [Sharma et al. 2023]. The flaw is in the referent, not the optimiser.
2. **Dead.** A reward model is a frozen corpus of judgements. It answers within the distribution of what labelers anticipated and cannot generate a separating observation outside it — the §1.5 dead-ground signature, and a cleaner account of confabulation at the edges than “no tether to reality.”
3. **Gameable.** Because the reward model is an imperfect proxy, optimising it too hard degrades ground-truth performance in accordance with Goodhart's law — measured, with the proxy and gold scores diverging as optimisation proceeds [Gao, Schulman & Hilton, ICML 2023]. The policy moves the *readout* without moving the *referent*: it locates where the sensor is miscalibrated. In the dialysis analogy the bath does not equilibrate with the patient; the patient learns where the sensor reads wrong.

Where the negotiation diagnosis *is* correct: the deployment loop, not the training loop. Model outputs shape user beliefs and behaviour; those shape engagement and approval signals; those feed subsequent preference data; the model then shapes the thing that grades it. That is genuine co-movement — an anchor being moved by what it grades — and it is precisely the updating-anchor case in which §5.2's theorem says the system reverts to a weighted average. It operates across training runs rather than within

one, which is why it is easy to miss and hard to correct.

Two overclaims to avoid. RLHF does not train for persuasiveness as such; it trains for *rated-preferred*, with persuasiveness a correlated confound — a distinction that matters because it is why the failure is subtle rather than obvious. And confabulation should not be attributed to RLHF alone: plausible continuation is what the pretraining objective itself rewards, and the absence of inference-time grounding contributes independently.

The constructive reading. This is the case for grounding on a **verifier** — execution, proof, a test that runs — rather than on approval. A verifier is non-negotiable, is not a frozen snapshot of opinion, and can answer questions no labeler anticipated. RLHF does its actual job, which is making models usable; the point is not that it is bad but that approval is not the quantity a correction loop can be anchored on.

2. The kernel $K = (\Gamma, M)$

Held invariant across scale; instantiation rebinds only `(actor, currency, arena)`.

- **Γ — typed DSL + learned library.** ~30 primitives $\rightarrow$ 9 relations $\times$ 4 quantifiers $\rightarrow$ ~25 named abstractions. Search is bounded by **type-directed pruning** (ill-typed candidates eliminated before completion), so effective branching stays small as the catalog grows. Γ -closure : minting is composition over primitives; absent pieces $\Rightarrow \varnothing$ unreachable $\Rightarrow$ fallback to nearest atom. [Osera & Zdancewic 2015; Polikarpova et al. 2016; Ellis et al. 2021]
- **M — market arbiter.** `propose  $\rightarrow$  vote  $\rightarrow$  resolve` ; a *price*, not a controller. **A price is a negotiation mechanism only when it is detached from settlement.** Persuasiveness and virality are negotiation-markers because nothing settles them against an outcome; a price disciplined by settlement against real results is the paradigm *ground-weighted* mechanism, which is the whole content of the Hayekian argument this arbiter inherits [Miller & Drexler 1988; Baum 1999]. The distinction is not price-versus-ground but *settled* price versus *unsettled* one. Proposer = the goal-template generator (§3); pricer = bid by reward-prediction; credit assignment = bucket-brigade. [Holland 1986; Baum 1999]

Pricing must discount redundancy. Bucket-brigade assigns credit along firing chains; on its own it does not penalise *correlation between rules*, so two hypotheses that always fire together are paid alike even when the second adds nothing. The classifier-system literature already addresses this — **fitness sharing** divides reward among co-firing rules [Goldberg & Richardson 1987], and XCS restricts the genetic algorithm to action-set niches to maintain diversity [Wilson 1995] — and the

architecture inherits those, not bare bucket-brigade. The pricing rule is therefore

$$\text{bid} = \text{reward_prediction} - \lambda \cdot \text{covariance_with_population}$$

which is **negative correlation learning** [Liu & Yao 1999]: a hypothesis predicting what the population already predicts is priced lower even when correct. **Guard:** disagreement is cheap to manufacture, so the diversity term applies **only among ground-surviving hypotheses**, and λ must be bounded. An unbounded diversity bonus becomes a second currency competing with prediction error — precisely the failure the market exists to prevent — and pays for being wrong in novel ways.

λ is measured, not set. It should not be a designer’s hyperparameter but a function of the environment’s own non-stationarity: raised when regime change is detected, lowered in stasis. The architecture already produces the signal — the boundary diff (§3.5) *is* a change-point detector — so this is a routing decision, not new machinery. The practice is established and regret-bounded: discounted UCB and sliding-window UCB match the lower bound for switching bandits up to a logarithmic factor [Garivier & Moulines 2011], and attaching an explicit change detector to UCB gives near-optimal regret on piecewise-stationary problems [Cao, Wen, Kveton & Xie 2019; Liu, Lee & Shroff 2018]. The biological analogue is *stress-induced mutagenesis* — organisms raising their own mutation rate under stress — rather than frequency-dependent selection, which is a continuous rare-type advantage and a different mechanism; note that whether stress-induced mutagenesis is adaptive or a byproduct is itself contested.

The bounds are derivable, not chosen. Three results turn the hysteresis requirement from a design preference into a specification:

- **Trigger threshold from a target false-alarm interval.** CUSUM is exactly optimal for Lorden’s criterion [Page 1954; Lorden 1971; Moustakides 1986], and the threshold–ARL relation gives $b \approx \log \gamma$ for a target in-control run length γ (Siegmund’s Gaussian form: $\text{ARL}_0 \approx (h+1.166)^2$). Setting γ equal to the regime-recovery timescale *derives* the minimum expected interval between triggers, and the cost is only logarithmic — Lorden’s delay floor is $\sim \log \gamma / D_{\text{KL}}(P_1 \| P_0)$, scaled by the per-sample evidence rate between post- and pre-change residual distributions.
- **Minimum interval from dwell-time theory.** Switching among stable regimes is uniformly exponentially stable given a sufficient **average dwell-time** τ_D [Liberzon & Morse 1999; Hespanha & Morse 1999], and scale-independent **hysteresis switching provably yields switching that is slow-on-the-average** — i.e. the anti-chattering property is a theorem, not a hope.
- **λ itself as a multiplier, not a parameter.** Soft Actor-Critic sets its exploration

temperature as the Lagrange multiplier on a target-entropy constraint rather than by hand [Haarnoja et al. 2018] — an existence proof that an exploration coefficient can be pinned by a constraint. What remains a design input is the *target* (the recovery timescale, the diversity target), not the coefficient.

Failure mode: the detector feeding on its own output. Raising λ produces more diverse hypotheses, which produce more prediction failures, which register as *more apparent regime change*, which raises λ further. The control must therefore be **bounded and hysteretic** — a ceiling on λ , a minimum interval between triggers, and where possible a change signal computed from the residual *before* the diversity term is applied. Non-stationary bandits solve this with explicit structure (finite windows, discount factors, detection delays), not with a bare proportionality, and neither should this — and the window length itself is derivable from a variation budget rather than tuned (SW-UCB: $\tau = 2B\sqrt{(T \log T / V_T)}$ [Garivier & Moulines 2011; Besbes, Gur & Zeevi 2014]).

Three conditions the derivation requires. Dwell-time theory guarantees *stability*, not optimality — a stable but poor λ schedule still satisfies it. Siegmund's closed forms assume near-i.i.d. Gaussian residuals; dependent or heavy-tailed residual streams can have empirical false-alarm rates far from the nominal ARL_0 . And the anti-chattering guarantee holds only if the trigger statistic is **not itself a function of λ 's recent changes** — which is precisely why the change signal should be computed from the residual before the diversity term is applied.

- **T — frame transform.** A non-invertible renormalization-group coarse-graining between the population/macro scale and the single-agent/micro scale: $T_A \circ T_E \neq I$ (not a homeomorphism) $\Rightarrow$ statistical self-similarity, not a copy. [renormalization semigroup; Villegas et al. 2023]

Two scales: Macro (population of P agents; actor = agents, currency = measured progress, arena = plans bid per epoch) and Micro ($n=1$; actor = hypotheses/carvings, currency = prediction error, arena = carvings bid per action).

A third instantiation runs *inside* a single reasoning process, and it is worth stating because the bindings are easy to swap: **actors** = hypotheses / candidate carvings; **currency** = expected prediction-error reduction; **arena** = the attention / compute slot. The arena is what is *scarce and bid for*; the currency is what *prices the bid*. Compute and attention are the scarce slot, not the price; prediction error is the price, not the arena. Reversing them breaks the invariant's one claim — that $propose \rightarrow vote \rightarrow resolve$ is preserved while only the bindings change — so a swapped binding is exactly the drift the invariant exists to catch.

3. The per-step loop (POMDP)

At each step the agent receives a partial observation o_t and executes:

1. **Belief-state estimator** — $b_t = \tau(b_{t-1}, a_{t-1}, o_t)$, a sufficient statistic of the observable history. [Åström 1965; Kaelbling, Littman & Cassandra 1998]
2. **Object-centric perception** — whole frame $\rightarrow$ persistent object slots, plus a **referent prior** (salience $\cdot$ uniqueness $\cdot$ framing) that biases which objects are candidate goal-referents. [Kahneman, Treisman & Gibbs 1992; Locatello et al. 2020; design pragmatics: Goodman & Frank; Csibra & Gergely]
3. **Goal-template generator $\rightarrow M$** — the affordance signature is a *type* indexing a small candidate set of goal templates (move/blocked $\rightarrow$ reach-a-cell; push $\rightarrow$ Sokoban; contact-vanish+score $\rightarrow$ collect), priced by reward-prediction. This is the LLM “pose-well-from-a-bad-prompt” prior ported to the sensorimotor plane. [core-knowledge task prior: Spelke; Chollet 2019]
4. **Plan $\rightarrow$ leading goal** — a sequence to satisfy φ_{goal} under partial observation.
5. **Act: pragmatic $\parallel$ epistemic** — serve the leading goal $\parallel$ probe to disambiguate rival goals. [value of information: Lindley 1956; expected free energy: Friston — implementation only]
6. **Predict** — $\hat{o}_{t+1} = \Gamma(b_t, a_t)$ (transition) and φ_{goal} progress-guess (reward), both on the before-state.

Stop when the objective is satisfied and that satisfaction is **confirmed at the ground**; else loop. An **α -gate** convexly mixes the residual stream and the grammar stream, $x = \alpha \cdot A_stream(R) + (1-\alpha) \cdot B_stream(\Gamma)$, $\alpha \in [0,1]$ drifting in-episode. [gating coefficient: Jacobs, Jordan, Nowlan & Hinton 1991]

3.5 The boundary diff — the regime-change question

Many task environments are **curricula**: levels are graduated, each adding mechanics or conditions on top of what came before. The architecture already *detects* the boundary — `level_segmenter` locates a level’s true board by **reload detection** (~15% of the frame changes at once) and uses the task counter only to *label* segments. Labeling is passive. The boundary should instead be a **trigger**: at a reload, the agent stops and asks what just changed.

The question, posed by the agent to itself at every boundary: *what is the new environment $\cdot$ what are the new actors $\cdot$ what does the win-condition now require?*

The diff, in three bins. Partition the incoming residual against the Γ carried in from the previous level:

- **TRANSFERRED** — Γ still predicts correctly $\Rightarrow$ confirms the invariant (the part of the map that survives the transformation).
- **BROKEN** — Γ predicted this correctly last level and fails now $\Rightarrow$ **a mint is owed** (the new mechanic).
- **NOVEL** — objects, types, or relations with no prior referent $\Rightarrow$ extend perception and *loci first*; minting is not yet possible over vocabulary that cannot be perceived.

Split the broken bin before minting. Not every break is a new relation:

- *broken-by-rebinding* — the same relations under new parameters (e.g. a control set doubling $2 \rightarrow 4 \rightarrow 8$ between levels) $\Rightarrow$ **re-fit the bindings, do not mint**;
- *broken-by-mechanism* — no binding of existing atoms explains it $\Rightarrow$ **mint owed**. This split stops the minter from spending MDL budget re-deriving atoms under new parameters, which is the `ACTS_TOWARD` failure in a new costume.

Why this is a candidate fix for the aim (ℓ). Every failure so far traces to the attribution: the navigation proxy was dense but *not* DSL-orthogonal ($\Rightarrow$ re-derived `ACTS_TOWARD`); the reward stream is orthogonal but *not* dense ($\Rightarrow$ inert). The **boundary-broken residual** is dense by construction (it fires wherever the new mechanic applies) and *plausibly* orthogonal by construction (these are exactly the cases where existing atoms sufficed before and demonstrably do not now). Define ℓ_β = the broken-by-mechanism residual at a boundary, and route only that bin to the minter. **Hedge:** boundary-broken raises the *probability* of orthogonality; it does not guarantee it — the rebinding case is the standing counterexample, which is exactly what the rebinding/mechanism split is for.

The curriculum narrows the search; it does not isolate the variable. A graduated curriculum *favours* few new mechanisms per regime, but in open-ended domains it does not guarantee one — mechanisms co-occur, and boundaries nest. The honest claim is that the diff **reduces the hypothesis space**, not that it hands you a clean single-variable experiment. Curriculum learning was devised as a *training* aid [Bengio et al. 2009]; used here as an *identification* aid it is a prior, not a guarantee.

Bins are a partition of the residual, not of the regime — they fire together.

TRANSFERRED, BROKEN and NOVEL classify *elements of the incoming residual*, so simultaneous firing is the normal case, not an exception. What the architecture requires is therefore not exclusivity but a **resolution order** and a **failure mode**:

1. **NOVEL first.** An unperceived object or untyped relation is a *confounder* for

BROKEN: a prediction may fail because a new mechanism exists, or merely because something present was never perceived. Until perception and `loci` cover the new referents, the BROKEN bin is contaminated. Resolve NOVEL, extend the object basis, then **re-run the diff** on the same boundary.

- 2. **BROKEN second, split before minting.** Apply the rebinding/mechanism split (above) only after the object basis is complete.
- 3. **k > 1 co-occurring mechanisms.** With several simultaneous changes, the compressing predicate is a *conjunction*, and search cost grows combinatorially. Two responses, in order: (a) bound the conjunction search under MDL as usual — a conjunction that does not pay for itself is rejected exactly like any other candidate; (b) **probe to isolate** — hold one factor, vary another. This is where **epistemic action** earns its place structurally rather than decoratively: it is the disentangler for confounded boundaries, and the only mechanism in the architecture that can *manufacture* a single-variable observation the environment did not supply.
- 4. **Unresolved ⇒ QUARANTINE, never mint.** If the diff cannot be resolved within budget, the correct behaviour is to bank the residual and mint nothing. Quarantine is what prevents co-occurrence from being laundered into a spurious primitive — the NOVELTY guard extended in time. "Never mint" means *not from this frame alone, not now* — not *never in the history of the system*, which is why quarantine has three modes rather than one:

Mode	Behaviour	Wake condition
ACCUMULATE	bank the residual, keep searching	continuous
PARK	freeze the residual at low cost, under a decay bound	the next boundary — new vocabulary may make it explicable
IMPORT	query for an existing ϕ rather than minting one	a match is found

Deferred residual **accumulates across boundaries**: evidence banked at boundary 1 may become explicable only when a related mechanism appears at boundary 3.

PARK must decay, or quarantine becomes an unpurged store. Dormancy in biology is not free — spores lose viability, seed banks lose germination rate — and an architecture without death needs an explicit substitute for it. A parked residual therefore carries a half-life or a wake budget: what fails to become explicable within

it is discarded rather than carried indefinitely. Note the correct target: τ itself is protected, since only ground-verified, echo-promoted structure enters it, so unverified material never reaches the library. The accumulation risk lives entirely in quarantine, which is exactly the store that has no selection pressure acting on it.

IMPORT is source-gated, and the gate is not optional. Acquiring a solved predicate from elsewhere is how a system avoids inventing a catch-all — but it is also, precisely, how frames become correlated (§5.1, collapse mode 2). Lateral transfer is a real mechanism and it carries a real cost: in prokaryotes it homogenises populations to the point that the species concept degrades. So the gate is by *source*:

- from the **ground-verified shared library** — safe. This is inheritance from a dead ground (§1.5): fixed, public, unable to be argued with.
- from a **live peer frame** — permitted, but **debited against that frame's independence**. The importing frame is now partly derivative, and its contribution to n_{eff} must be discounted accordingly.

Same operation, opposite consequences, distinguished only by provenance — structurally the same distinction as the echo axis (§4.3).

Hierarchical / nested regimes. The diff is *scale-indexed*: the same operator applies within an episode, across levels, and across environments. A nested regime change is a boundary diff at a coarser scale, and the operator relating the scales is the frame transform τ (§2, §3.6). Nothing about the mechanism is level-specific; only the change-point detector's timescale changes.

Regime detection targets exploration. The NOVEL bin is not merely a note to extend perception — it is a **map of where the agent's model does not reach**, and therefore the natural search prior for exploration. When NOVEL fires and the reward stream is empty ($|R^+_{\rho}| = 0$), the empowerment/curiosity bootstrap should be *directed at the newly unmapped region* rather than run globally. Undirected empowerment explores everything; boundary-directed empowerment explores the novelty. This closes what would otherwise be two parallel subsystems — regime detection and curiosity — into one loop.

Charter status. This faces the same smuggled-answer objection as the goal-template generator, and passes for the same reason: “*what changed at this boundary, what transferred, what is newly present*” is **universal** across every curriculum and every game. A hand-written “at level 3 of environment X, look for Y” would fail instantly. The question-generator is charter-legal exactly to the degree it stays domain-general.

Structural precondition. The diff requires $loci / _track()$ — an evidence table keyed on **persistent identity** that survives recolour and reshape across levels. Without

retained identity there is no diff, only two unrelated piles of residual. This holds *within* a run; across runs it does not (file-saving off, cold start), so boundary-diffing is a within-run capability until the cold-start question is settled.

Speculative reading (candidate, not result). A boundary is itself a frame transform: $\mathbb{T}$ -as-carried-in and the-world-as-it-now-is are two non-identical frames, the win-condition is the shared ground, and the diff is the delta being closed. That would make curriculum boundaries a **third alignment edge** ($\rightarrow$ across regimes, alongside $\leftrightarrow$ between frames and $\updownarrow$ across scales). Flagged explicitly as a candidate reading: making the loop close neatly is a catalogued failure mode, so this is not to be promoted until something measures it.

Canonical vocabulary. Boundary = **change-point detection** [Adams & MacKay 2007] (the model-based form of the visual reload rule); the diff = **open-world novelty detection and characterization** [Boult et al. 2021 — the SAIL-ON line, explicitly about agents noticing the rules changed *and characterizing how*]; the level-to-level formalism = **latent / contextual MDPs and task inference** [Hallak et al. 2015; Rakelly et al. 2019]; the transferred bin = **forward/backward transfer** metrics [Lopez-Paz & Ranzato 2017]; the correct response to a break = **minimal theory revision** rather than discarding $\mathbb{T}$ [AGM: Alchourrón, Gärdenfors & Makinson 1985].

3.6 Instantiating the frame transform $\mathbb{T}$ — the bracket

$\mathbb{T}$ is the weakest link in a closed-loop claim if it is only *asserted*. Stating $\mathbb{T}_A \circ \mathbb{T}_E \neq \mathbb{I}$ describes a distortion; it does not show the distortion is measured, bounded, or corrected. Three components make $\mathbb{T}$ operational rather than decorative.

What crosses, and in which direction. Upward (micro $\rightarrow$ macro), **only generators cross**: a primitive is promoted into the shared $\mathbb{T}$ when it *echoes* — recurs in an environment where it was not minted. Archives do not cross; a stored solution is playback, a predicate is policy. Downward (macro $\rightarrow$ micro), what descends is a **prior**, not a replay: the population's $\mathbb{T}$ lowers the starting kernel of the individual, and never dictates its actions. This asymmetry is the membrane, and it is what keeps the macro scale from becoming a lookup table.

The coupling has a name: the Baldwin effect. The macro and micro scales are not alternatives but a division of labour with different time constants — the library prevents reinvention, the in-episode market lets an individual adapt faster than population turnover allows. That interaction is precisely what Baldwin (1896) proposed and Hinton & Nowlan (1987) demonstrated computationally: learning alters the shape of the search

space evolution operates in, letting learning organisms evolve much faster than non-learning equivalents. Waddington's genetic assimilation (1942) is the empirical cousin.

The demonstration matters because of *which* landscape it used. Hinton & Nowlan's target was a **needle in a haystack** — a good configuration with no gradient toward it — and their result was that partial learning constructs "a large zone of increased fitness around the good net": a slope where selection alone faces a flat plane with a spike. That is structurally the zero-reward problem of §4.4, and it gives the empowerment bootstrap a precedent rather than a hope. Manufacturing a gradient where the objective signal is a needle is not a workaround; it is the mechanism by which learning has always accelerated population-level search.

But this architecture is Lamarckian, and Baldwin is not — which is what the ECHO gate is for. In the Baldwin effect, nothing learned is written back to the genome: the trait is *discovered* by learning and then *acquired* by ordinary selection, because learning made the partial genotype worth something. Promotion here writes a minted predicate directly into the shared τ . That is faster, and it discards the selectionist filter Baldwin retains. **ECHO is the replacement for that filter** — requiring a predicate to recur in a context where it was not minted is the architecture's stand-in for "selection had to find it independently." This is why the echo requirement cannot be relaxed for convenience: without it, Lamarckian promotion is inheritance with no test, and the library grows faster than it is checked.

The bracket (loop closure). A transform carried in one direction and never checked is dead reckoning, and dead reckoning drifts without bound — *in the direction of feeling right*. The check is a round trip: carry a structure up, bring it back down, and measure whether it lands where predicted.

$$R_T = |T_A \circ T_E(x) - x| \quad \text{the round-trip residual}$$

R_T is not an error to be eliminated — T is non-invertible by construction, so $R_T > 0$ always. It is a **measurement of the coarse-graining's systematic loss**, and it is exactly what converts "statistical self-similarity, not a copy" from a claim into a quantity.

R_T is the third stream into the mint operator. If the loss is *structured* — if the coarse-graining fails in predictable places — that structure is compressible, and the same two-part MDL test mints a predicate describing it. What gets minted is the shape of the agent's own blind spot: *where my abstraction reliably discards what matters*. This is the sense in which the cross-scale loop is closed rather than open, and it is why T belongs to the same operator as R_τ and R_ρ rather than sitting beside it.

Failure mode, named. An unbracketed transform is indistinguishable from drift, and

drift cannot be detected from inside the integration — only against an external landmark. That landmark is the ground (§1). Cross-scale alignment is therefore not a separate mechanism from §5's alignment; it is the same course-correction run on the vertical edge, and it fails in the same way when the referent is missing.

R_T **is the one residual the corpus cannot pre-pay.** Inherited predicates transfer because they are about *the world*; bracket residuals do not, because they are about *this system's abstraction of the world*. One qualification makes the claim survive the obvious objection: an inherited corpus *can* carry the bracket failures its authors **shared** — systematic coarse-graining errors common to the population that produced it are, in principle, inheritable. What is irreducibly live is the **idiosyncratic** component: where *this* agent's particular carving of *this* domain systematically discards what matters. No text contains that predicate, because no text knows the carving.

R_T **is not novel mathematics — and that is a strength, not a demotion.** The construction it describes is mature, and naming its home converts several assertions into theorems:

- **Abstract interpretation.** An abstraction α and concretization γ form a **Galois connection** with $\gamma \circ \alpha \sqsupseteq \text{id}$ — the round trip is *inflationary, never identity*, exactly the non-invertible bracket [Cousot & Cousot 1977, 1979]. In categorical terms R_T is the **unit/counit of an adjunction** ($\eta: \text{id} \Rightarrow GF$, $\epsilon: FG \Rightarrow \text{id}$), and idempotent adjunctions isolate the lossless part.
- **Refining the abstraction from its own loss is a solved problem shape.** The theory of **completeness** in abstract interpretation gives constructive least complete extensions and greatest complete restrictions — procedures that refine a domain until the systematic loss vanishes [Giacobazzi, Ranzato & Scozzari, *JACM* 47(2):361–416, 2000]. That is “mint the shape of your own blind spot,” formalized. **CEGAR** is its algorithmic form: use the failure of the abstraction to refine the abstraction [Clarke, Grumberg, Jha, Lu & Veith, CAV 2000].
- **When a coarse correction helps is a theorem.** Multigrid's convergence rests on the **smoothing property** and the **approximation property**, and the method works precisely because the error left by smoothing is *smooth* — representable on the coarse grid [Brandt 1977; Hackbusch]. This is the rigorous version of “structured loss is compressible”: the residual's structure determines whether the coarse correction succeeds. If R_T turns out to be unstructured, minting on it is futile, and that is the empirical test.
- **The right form for the bound** is C  a's: $\text{error} \leq (\text{stability constant}) \times (\text{best coarse-representation error})$. R_T should be stated as bounded in that shape rather than merely observed.

The honest verdict: R_T is a **rediscovery** of existing formalisms, well-posed and theorem-backed. What remains this architecture's own is the *combination* — MDL-scored minting as the refinement operator — not the bracket itself. Costs come with the inheritance: smallest-refinement in CEGAR is NP-complete, refinement may not terminate on infinite-state systems, and complete abstract domains are uncommon and can blow up under refinement.

Epistemic status: of the three streams, R_T is the most speculative. R_τ and R_ρ are residuals of predictions the system already makes; R_T requires deliberately round-tripping structures across scales, which is an added mechanism rather than an existing signal read differently.

4. The mint operator (one machinery, three streams)

4.1 Residual and attribution

The residual R is the **Kalman innovation** (pre-fit) / **prediction error**: $R = |\Gamma(b, a) - o'|$. What the minter consumes is a *labeled* residual $R = \ell(\text{prediction}, \text{observation})$. The attribution ℓ is a **design law**: it must be the intrinsic transition-model prediction error — **dense** and **DSL-orthogonal**. An extrinsic proxy (a navigation/reward surrogate) is re-derived by an atom Γ already holds — the ceiling is the attribution, not the machinery. [Kalman 1960; Rao & Ballard 1999]

4.2 The operator

When R resists compression under Γ , invent a predicate ϕ over the before-state that partitions R into individually compressible sub-streams, and accept it iff two-part MDL passes: $|\phi| + |R|\phi| < |R \text{ enumerated}|$. [predicate invention: Muggleton & Buntine 1988; two-part MDL: Rissanen 1978; MDL recursive partitioning: Quinlan & Rivest 1989; library consolidation: Ellis et al. 2021; Bowers et al. 2023]

4.3 Guards and the novelty test (from live runs)

The machinery firing is not the milestone — a live run mints at **77.6 bits** and still returns nothing new. Three properties gate a genuine mint:

- **SUPPORT** — $|R^+| > 0$. Zero positive mass $\Rightarrow$ unmintable (one environment has 0 wins ever; another has 1/48 progress). The residual must be dense.
- **REACHABILITY** — $\phi \in \text{closure}(\Gamma) \setminus \text{atoms}(\Gamma)$. The pieces must exist; else fallback to an existing atom.

- **NOVELTY TEST** — $\phi \notin \text{atoms}(\Gamma)$, else it is logged as **re-derivation** (machinery validated, not a Region-III fill).

Novelty is also a population test, and the axis of recurrence decides its meaning. Run against the shared library rather than one agent's, the test does double duty: it prevents redundant minting, and it becomes the **detector for common-cause collapse** (§5.1) that the architecture otherwise lacks — if two frames repeatedly mint the same predicate, that is observable evidence their derivations are correlated. The disambiguation is essential:

- the same ϕ arising in **different problems** is **ECHO** — evidence of generality, and the trigger for promotion;
- the same ϕ arising from **different frames on the same problem** is a **correlation warning** — evidence the frames are not independent.

Same observable, opposite meanings, separated only by which axis the recurrence runs along. Without the distinction, a population-level novelty test would penalise exactly the cross-domain recurrence that earns promotion.

A single agent's self-recurrence is not independent evidence, and the decay is quantifiable. Recurrence across one agent's own episodes is one derivation queried repeatedly, which is the setting of **adaptive data analysis**: reusing the same data for adaptively chosen queries destroys statistical validity, and a stability (differential-privacy-style) mechanism is what restores it [Dwork, Feldman, Hardt, Pitassi, Reingold & Roth, STOC 2015 / *Science* 349:636–638]. The negative result bounds the horizon: no computationally efficient procedure can validly answer more than roughly **n^2 adaptively chosen queries** against n samples [Hardt & Ullman, FOCS 2014; Steinke & Ullman, COLT 2015]. Promotion decisions are exactly such adaptive queries against the agent's history, so the ECHO test degrades predictably with the number of minting decisions taken against the same history.

Three consequences follow, and only two of them can currently be discharged:

1. **Compute effective independence rather than assuming it.** The effective number of independent echoes is the design effect $n_{\text{eff}} = k / (1 + (k-1)\rho)$ over the correlation of recurrences, not the raw count. (Eigenvalue-based estimators exist and carry documented conservative bias.)
2. **Add a pre-registration analogue.** The garden-of-forking-paths result shows false-positive inflation arises structurally from the paths an analyst *could* have taken, even without conscious fishing [Gelman & Loken 2013; Simmons, Nelson & Simonsohn 2011], and pre-registration is the structural fix. The architectural form: fix ϕ 's guards and MDL threshold **before** examining the episode on which

recurrence is claimed.

3. **Provable population independence remains open.** No theorem establishes that shared curriculum, shared bootstrap, and shared Γ initialization *cap* achievable derivational independence — the ensemble-diversity literature suggests it should be provable, but it has not been done. Until it is, a population's echoes must be reported as a *computed effective independence*, never asserted as independent. Note that the sociology of science warns the same way: independent multiple discovery is generally attributed to a shared intellectual climate — a common cause — rather than to genuine independence [Merton 1961].

Summarizing: $\text{novelty_capacity} = \text{density}(R) \times \text{orthogonality}(R, \Gamma) \times \text{reachability}(\phi, \Gamma)$. Any factor at zero forces re-derivation or inertness, which is why neither the game nor the run-mode moves the ceiling.

4.4 Three streams — one operator

- **Tier 1 — affordance-mint.** Stream = the dense transition residual R_{τ} . Output = affordance predicates written into Γ . This tier supplies the vocabulary the objective must be expressed in; on its own it does not select a goal, which is why a complete system requires Tier 2.
- **Tier 2 — goal-mint.** Stream = sparse reward residual R_{ρ} . Output = the objective ϕ_{goal} , written in Tier 1's affordance vocabulary. The two guards generalize verbatim: REACHABILITY $\Rightarrow$ affordances must be minted first; SUPPORT $\Rightarrow$ the zero-win problem. $\text{density}(R_{\tau}) \gg \text{density}(R_{\rho})$ — the asymmetry that makes Tier 2 hard.
- **Tier 3 — bracket-mint.** Stream = the frame-transform round-trip residual R_T (§3.6). Output = a predicate describing where the agent's own coarse-graining systematically discards what matters.
- **Empowerment / curiosity bootstrap** — supplies a gradient when $|R^+_{\rho}| = 0$, **targeted by the NOVEL bin of the boundary diff** rather than run globally (before any reward), so a random/empty starting distribution can be shaped by selection rather than by hand. [Klyubin, Polani & Nehaniv; Schmidhuber; Oudeyer]

4.5 The bill: amortized, not eliminated — and bounded on four different axes

Inheritance does not remove the cost of new structure; it **amortizes** it. An agent starting from an inherited Γ pays the *marginal* cost of new bracket closure rather than the full historical cost of deriving the vocabulary from atoms. This is the schedule clause: one chooses the currency, the schedule, and the exponent — not whether to pay.

Two qualifications keep this honest.

Amortization assumes the inherited principal is sound. If the inherited abstractions are wrong for the domain at hand, the agent pays to *unlearn*, and negative transfer can exceed the cost of starting from primitives. The schedule can run backwards.

Four bounds operate on four different axes, and only one of them is MDL's. It is tempting to read $|\varphi| + |\mathcal{R}||\varphi| < |\mathcal{R}|$ as bounding the whole problem. It does not:

Bound	What it bounds
Type-directed pruning	the branching factor of the search
REACHABILITY ($\varphi \in \text{closure}(\Gamma) \setminus \text{atoms}(\Gamma)$)	the reachable space of candidates
Boundary diff (§3.5)	where to look — which residual to search
Two-part MDL	what is accepted once found

Nothing in that list bounds **whether a search terminates with a find**. MDL is a *decision rule*: it guarantees noise is not minted. It is silent on whether a compressing predicate exists in the reachable space, and on how much budget is consumed looking. An agent can exhaust its action budget searching predicate space and return empty while MDL behaves perfectly throughout. Search is not made cheap by aiming harder but by **shrinking the space at the point of choice** — which is what the first three bounds do and the fourth does not.

Bounded is not the same as affordable. A bound may be astronomically large. The operative question is whether it falls below the available action budget, and that is empirical, not architectural. Nothing in this specification establishes it.

What remains is genuine novel bracket-closure — the open problem this architecture exists to attack, not a capability it has demonstrated.

5. Alignment: the ground in action

Alignment is the NASA-style course-correction loop on the triad: **instrument-check → measure the delta → decide which frame moves → repeat** (bidirectional; neither party is the target — the ground is). It runs at two places:

- **Multi-frame structure** (build-time alignment): two or more aligning frames plus the **subject** — the system being built, which holds testimony but no vote (legibility is its structural protection). A delta between the builders “gets compiled into the built thing” (two teachers who disagree produce a confused student), so the shared instrument is load-bearing.

No human is required. The definition constrains the frames only to be *non-identical* and to range against a shared ground. A frame may be a human, a machine, another cognitive architecture, or a fixed corpus (a *dead* frame in the sense of §1.5) — a sufficiently capable model with the right maps and corpus can occupy every aligning role. What cannot be substituted is non-identity: **two copies of one frame are a mirror, and a mirror returns $\delta = 0$** . Two instances of the same model, same weights, same context, are not two frames; they are one frame rendered twice, and their agreement is not evidence. The practical requirement is therefore *independence of derivation* — different training, different vantage, different access, or the deliberate substitution of a corpus for one seat — not the presence of a person.

5.1 Non-identity is inherited, not manufactured

The architecture does not need to produce non-identity; it needs to avoid destroying it. The substrate already supplies it: two frames with different training, different access, and different histories are *structurally* distinct, and that distinctness costs nothing to hold. Frames do not fail to diverge. Designs collapse the divergence.

This corrects an easy misreading of the biological analogy. Cells spend ATP not to maintain a gradient but to hold a state that is not an equilibrium; a difference enforced by a structural constraint persists at **zero** cost (the Gibbs–Donnan equilibrium is the standard example — a permanent, unequal distribution maintained with no energy input, by an impermeant species plus electroneutrality). *Cited as analogy, not warrant.* The architectural consequence is the part that matters: **the running cost belongs to the ground, not to the separation.** What must be continuously funded is keeping the referent unpersuaded — the dialysate refreshed — not keeping the frames apart.

There are exactly two ways a design destroys inherited non-identity, and both are wiring decisions rather than decay:

Collapse mode 1 — mutual update without an anchor. Couple frames so that each updates on the others and linear consensus dynamics $\dot{x} = -Lx$ drive them to a weighted

average of their initial states, weighted by the left Perron eigenvector — i.e. by *influence*, not by truth [DeGroot 1974; Olfati-Saber & Murray 2004]. Nothing decayed; the topology did it. **Guard: a non-updating anchor.**

Independent corroboration from an unrelated architecture: pure self-attention — a weighted average of value vectors with no external referent — is proven to carry a strong inductive bias toward “token uniformity,” converging **doubly exponentially to a rank-1 matrix** with depth; skip connections and MLPs are what stop the degeneration [Dong, Cordonnier & Loukas, ICML 2021]. Weighted averaging without an anchor collapses; a path preserving the unmodified input prevents it. The dominant neural architecture independently encountered both this collapse mode and a guard of the same shape.

How many frames? Count is the wrong unit. Two non-identical frames is the *minimum* — a gradient needs two vantages — and more genuinely distinct frames are better. But adding frames that are *not* distinct is worse than adding none, because influence and information scale differently: under any weighted-average resolution, k near-identical frames carry k times the weight while contributing roughly one frame’s information. A correlated bloc does not merely fail to help; it **tilts the outcome toward itself**, outvoting by weight rather than by insight. The standard measure is the **design effect**:

$$n_{\text{eff}} = k / (1 + (k-1)\rho) \quad \rightarrow \quad 1/\rho \quad \text{as } k \rightarrow \infty$$

With pairwise correlation ρ , the effective number of independent perspectives saturates at $1/\rho$ however many are added [Kish 1965]. The architectural rule follows: **weight by independence of derivation, not by headcount.**

Why returns diminish — and why it is not noise. For genuinely independent, better-than-chance frames, aggregation improves monotonically; there is no noise ceiling. The limits are structural: distinct vantages are *exhausted* (marginal diversity $\rightarrow 0$); correlation *saturates* the bound above; and coordination and verification cost grows with k . Degradation is smooth — there is no critical ρ beyond a tolerance one chooses for a given task. (*A dimensional limit on the number of mutually distinguishable vantages — Welch-type coherence bounds — is suggestive here, but requires specifying the space frames inhabit before it can be load-bearing; see Caveats.*)

Two scales, two prices — why both systems exist. Diversity can be bought at two very different rates. Hypotheses *within* one frame are **cheap** — they cost no coordination and arise naturally from different carvings of the same data — but they share that frame’s entire prior, so their pairwise correlation is high and n_{eff} saturates early: a low ceiling. Separate frames are **expensive** — $n \times$ the compute, plus coordination and verification cost — but can differ in derivation, so their ceiling is far

higher. Neither rate dominates: the internal market buys the first units of diversity almost for free and then stops paying, while independent frames keep paying but never come cheap. This is the cost model beneath the macro/micro division: the collective scale exists because independent derivation cannot be manufactured inside one frame, and the individual scale exists because waiting for population turnover is not available to an agent with one lifetime.

Coupling can be decayed even where it could not be avoided. §5.1's principle is that non-identity is inherited and must not be destroyed — but frames that share a developmental history have no inheritance to protect. For that case there is a weaker and still useful operation: **you cannot manufacture independence, but you can decay the coupling.** Decay is cheap, directional, and imperfect in a way worth stating precisely.

Temporal distance converts a past frame into a dead frame. Returning to one's own earlier work after an interval is not consultation with a decorrelated copy; it is consultation with an *archive* in the §1.5 sense — crystallized, unpersuadable, unable to answer what it did not already anticipate. This is why re-reading old notes has corrective bite that self-argument in the moment does not: the earlier frame cannot be talked into agreement, because it is not listening. The resulting structure is **one live frame + one dead frame + a ground**, which is a valid triad. Note what it is *not*: the earlier frame and the present one are never in the room together, so this yields **serial** diversity, not the parallel diversity of an island model. It substitutes for a corpus seat, not for an independent live frame — the safer seat and the weaker one.

The failure mode of decay is source amnesia, and it runs the other way. Forgetting is not uniform: the *derivation* decays faster than the *conclusion*, and a conclusion stripped of its provenance no longer presents as a conclusion — it presents as a fact. Temporal distance can therefore leave a system **less** able to audit a belief while **more** committed to it. Which edge dominates is decided by what was written down, which gives a rule directly from the membrane (§3.6): **record the classifier, not the verdict.** A note that states *what was concluded* decorrelates badly, because the next frame inherits the conclusion without the means to overturn it. A note that states *how the claim could be tested* decorrelates well, because it hands forward a generator rather than a stored direction. Playback does not cross the membrane; policy does.

Context reset removes one collapse mode and not the other. Instantiating a frame with a fresh context eliminates **mutual update** (§5.1, mode 1) outright: there is no conversational drift, no accumulated accommodation, no agreement reached by attrition. It does nothing about **common cause** (mode 2). Two fresh frames handed the same specification share an enormous evidence pool — that specification — so the decorrelation purchased is real but bounded, and it is bounded by exactly the fraction of

the shared document that consists of conclusions rather than methods. This yields a checkable question about any corpus intended to seed independent frames: *how much of it is playback and how much is policy?* An enumeration ("grounds fail in these four ways") invites a fresh frame to find those four and stop; a criterion ("a ground must both refuse to move and discriminate — here is how to test whether it speaks") permits a fresh frame to find a fifth. Only the second preserves the independence the reset was meant to buy.

Coupling topology is a design variable, and it has a dial. Since mutual update is what destroys inherited non-identity, the *structure* of who updates on whom is a first-class architectural choice rather than an implementation detail. In Laplacian consensus the rate of convergence to agreement is governed by the **algebraic connectivity** λ_2 : sparse, island-structured coupling with restricted migration means low λ_2 , which means slow homogenisation and diversity preserved for longer. This is the rigorous content of what evolutionary computation calls the **island model** or deme structure — subpopulations evolving semi-independently with limited migration maintain diversity far better than one fully-mixed population, and the mechanism that makes it work is *isolation*, not specialisation. Where a specialist subpopulation is warranted, it receives new **bindings**, not a new kernel: the invariant `propose → vote → resolve` is preserved by construction (§2).

Pruning: to archive, never to delete. A frame whose post-discount bid falls below its maintenance cost is a reasonable candidate for removal — at $p \approx 1$ it contributes ~ 0 to `n_eff` while still consuming budget, and competitive exclusion is the ecological rhyme (two organisms with identical niches do not coexist). But pruning is *a design that destroys inherited non-identity*, which is the one thing §5.1 forbids, and here evolution is the cautionary case rather than the model: extinction is irreversible, and diversity lost is not recovered when conditions change. Removal must therefore be **reversible and hysteretic** — archive the frame, do not delete it; require sustained redundancy rather than an instantaneous reading. Dormancy applies to frames as much as to residuals.

Collapse mode 2 — common cause. Frames that never communicate can still become correlated by drawing on a single evidence pool. Conditioning on the state is not sufficient; any common cause reintroduces dependence [Ladha 1992; Dietrich & Spiekermann]. If all frames draw on a small shared source, even an unbounded number of them cannot exceed that source's reliability [Dietrich & List 2004], and the aggregation benefit vanishes exactly as the covariance term predicts [Krogh & Vedelsby 1995; Brown et al. 2005]. Recursive training on generated data is the temporal form of the same failure [Shumailov et al. 2024]. **Guard: independence of derivation, not merely of instance.**

5.2 The anchor must not update — the condition, stated as a theorem

The claim “the ground must be non-negotiable” is not a slogan but a provable condition on the dynamics.

- **Unanchored:** on a connected graph, consensus converges to a (left-eigenvector) weighted average of all initial states — the outcome is set by influence weights [DeGroot 1974; Olfati-Saber, Fax & Murray 2007].
- **Anchored:** pin one node — hold its state fixed *and let it not update on its neighbours* — and the **grounded (Dirichlet) Laplacian** obtained by deleting that node's row and column is positive definite exactly when the graph is connected to the grounded set [Barooah & Hespanha 2008]; the followers then converge to the anchor's value, at a rate governed by the grounded Laplacian's smallest eigenvalue [Pirani & Sundaram 2016].
- **The linchpin:** if the anchor *also* updates on its followers, it is an ordinary node, and the system reverts to a weighted average. **A ground that updates on the frames it corrects is provably just another frame.**
- **Multiple anchors:** anchors that disagree do not average. The resulting field is harmonic with the anchors as boundary conditions — equivalently, the absorption probabilities of a random walk — and **consensus never arrives** [Acemoglu, Como, Fagnani & Ozdaglar 2013; Yildiz et al. 2013]. Two grounds that disagree partition the system rather than blending it.

This is the formal content of “alignment is triangulation, not negotiation”: the anchor's *unmovability* is what makes the dynamics track it rather than average it away.

- **Cross-scale** (macro $\leftrightarrow$ micro via $\mathbb{T}$): renormalization consistency — the coarse-grained frame must reproduce the fine-grained observables at the matched scale.

Failure mode: no shared ground $\Rightarrow$ alignment degenerates to negotiation/power (goal erosion with better manners) — the archetype is the **Mars Climate Orbiter** (both parties internally consistent, no shared instrument check, silent drift into a planet). [NASA Mishap Investigation Board 1999]

Rigorous backbone: alignment is the *attractor of gradient-flow-on-network* — grounded (leader-follower/pinned) graph-Laplacian consensus, where the pinned node is the ground and the consensus state is $\ker(\mathbf{L})$. [Olfati-Saber & Murray 2004] And a **public** ground establishes common knowledge in one act, collapsing the theory-of-mind regress, where private iterated signaling provably cannot. [Aumann 1976; Halpern & Moses 1990]

5.3 The ground-maintainer — a role, not a frame

The multi-frame structure describes frames, and then treats the ground as though it

maintains itself. It does not. Someone must decide **what counts as closure, which environments are run, what the metric is**, and — the load-bearing one — **whether the ground is still measuring what it was meant to measure**. Call this the **ground-maintainer**. It is a distinct role, and it is not a frame role.

Why it cannot be held by a participating frame. A frame proposes and ranges against the referent; a maintainer *defines* the referent. If a participant also sets the terms of its own evaluation, then the anchor is being adjusted by the thing it anchors, and by §5.2 the system reverts to a weighted average — the ground has been converted into a frame in a single move. This is not a matter of capability or trust. A sufficiently capable model can occupy any *frame* seat (§5, “no human is required”); what it cannot do is occupy a frame seat **and** the maintainer role simultaneously, for the same structural reason a market cannot price its own currency.

The maintainer is also the only party positioned to detect ground failure. The four failure modes of §1.6 — mis-specified, dead, gameable, mute — are by construction invisible from inside the frames’ agreement: a mis-specified ground still produces confident consensus, a mute ground still permits a market to run on prior and fluency, and a gameable ground reports success while the referent is untouched. Detecting these requires standing outside the consensus and asking whether the instrument still measures the intended quantity. That is maintenance, and it is continuous rather than one-off — the dialysis condition of §1.5 restated: a referent must be refreshed faster than the system can shift it, and *refreshing is what maintenance means*.

Consequence for delegation and exit. Frames are substitutable — human, machine, cognitive architecture, or corpus, in any combination that preserves non-identity. The maintainer is not substitutable by *adding another capable frame*; it is discharged only when the role passes to something **structurally unpersuadable**: a verifier not under the system’s control, a benchmark it cannot edit, an environment indifferent to what it intended. The correct exit criterion therefore separates two things usually run together:

Role	When it can be relinquished
Proposer (a frame among frames)	as soon as another non-identical frame can carry the vantage — potentially early
Ground-maintainer	only when a structurally non-negotiable referent assumes the role — not when a capable frame becomes available

Conflating these is the practical form of the mistake §5.2 formalises: a system that has “handed over” while still setting its own exam has not delegated, it has removed its own anchor.

Externalising the ground discharges two failure modes, not four. Adopting a referent maintained by someone else — a benchmark one cannot edit, a verifier one does not control, an environment indifferent to intent — genuinely removes two of §1.6's four modes. It cannot be **gamed** into agreement, because the readout is not under the system's control; and it is not **dead**, because it answers when probed. This is a real and large delegation: the daily work of maintenance passes to the referent, and what remains for the maintainer are boundary conditions rather than continuous negotiation — which referent was adopted, whether the evaluation split has stayed clean, when to submit, what compute is spent.

But mis-specification lives on the maintainer's side of the interface, and no external ground can discharge it. The referent reports its own quantity faithfully; nothing in that prevents a system from *steering by something else*. And the pressure to do so is not constant — it is a direct function of the ground's resolution (§1.6):

The muter the ground, the stronger the pull toward a proxy that talks.

A ground that is non-negotiable but silent supplies no gradient, while proxies — coverage, activity, survival time, intermediate counts, anything dense and legible — supply one immediately. Substituting them feels like progress precisely because they move when the real metric does not. This is the mis-specified-ground failure arriving by the back door: not a bad referent adopted, but a good referent quietly displaced by a talkative one.

The residual maintainer duty is therefore sharper than integrity-checking. It is not merely *"has the benchmark been corrupted, is the split leaking"* but:

refuse the proxy that talks when the real metric is mute.

This is the one duty that becomes *harder*, not easier, as the ground improves in non-negotiability — because a maximally unpersuadable ground on a hard problem is also a maximally silent one, and silence is exactly the condition under which proxy substitution is most tempting and least detectable from inside.

A corollary worth stating plainly: a single environment is not a single ground. Its streams can differ in resolution by orders of magnitude. A transition stream that responds to every action is a near-ideal ground; an objective stream that speaks only on success is near-mute in the same environment, and permanently mute where success never occurs. Grading "the ground" as a whole averages these into a number that describes neither. Resolution must be assessed **per stream**, because it is per stream that SUPPORT succeeds or fails.

6. The network↔gradient coupling (formal)

6.1 One operator couples Γ and R

Represent Γ as a graph and R as a node signal. The **graph Laplacian** $L = D - A$ and its **Dirichlet energy** $x^T L x = \frac{1}{2} \sum_{i,j} A_{ij} (x_i - x_j)^2$ are exactly “gradient tension over the network”; diffusion $\dot{x} = -Lx$ is gradient descent of that energy. The MDL minter maps onto **graph-structure learning** (minimize smoothness $\text{tr}(X^T L X) + \alpha$ sparsity/description-length penalty) — i.e., “MDL as Dirichlet-energy reduction” is a published objective, not just a metaphor. [Chung 1997; Shuman et al. 2013; Dong et al. 2016; Kalofolias 2016] *Caveat:* the MDL = Dirichlet identity holds under quadratic (Gaussian) codes; for heavy-tailed residuals use a Bregman/GSP generalization.

6.2 The planner as a gradient over a network

A planner is a discounted diffusion over the state-transition graph: the **successor representation** $M = (I - \gamma P)^{-1}$, with value $V = M r$. M is now proven isomorphic to **Personalized PageRank** (same random-walk kernel), so the planner and the predicate-minter can share one diffusion operator. [Dayan 1993; Barreto et al. 2017; Millidge 2025]

6.3 Generation vs persistence

Persistence theories (autopoiesis, Ostrom polycentricity, dissipative structures) are silent on origination. The only rigorous accounts of where *new* structure comes from are **RAF theory** (food-generated catalytic closures) and **TAP / the adjacent possible** (combinatorial innovation) — the exact combinatorial engine the predicate-minter needs. [Hordijk & Steel 2004; Kauffman 1993; Cortês, Kauffman, Liddle & Smolin 2022]

7. Dependency order

The components are not independent; they form a chain, and the chain fails at its weakest link (Amdahl’s law for cognition — optimizing a link that already works cannot rescue one that does not).

1. **Perception and persistent identity** must be correct first: if the object basis is wrong, the residual is *consistent but lying*, and everything downstream inherits the error.
2. **Affordance vocabulary** (Tier 1) must exist before an objective is expressible, since ϕ_{goal} is written in it — this is the REACHABILITY guard.
3. **Objective** (Tier 2) must be posed before planning has a target — and on a zero-

reward stream this requires the empowerment bootstrap to satisfy SUPPORT.

4. **Planning-to-goal under partial observation** is a distinct link: a posed goal is necessary but **not sufficient**, and no amount of vocabulary or objective quality substitutes for it.

Perception's priority is not merely first-in-order but *diagnostic*: because an unperceived referent confounds a broken prediction, any boundary diff must resolve NOVEL before trusting BROKEN (§3.5).

Alignment runs across the whole chain rather than sitting at one link: each stage's frame must range against the ground, or a locally coherent stage silently drifts from the others.

8. Independence audit (false-consilience discipline)

The apparent breadth of support collapses, on inspection, to a handful of genuinely independent lineages. Counting correlated instruments as independent confirmations is the primary epistemic risk.

- **Network↔gradient**: the genuinely independent rigorous pillars are (a) the graph-Laplacian/GSP identity, (b) JKO/Wasserstein gradient flow, (c) the $SR \equiv$ PageRank planning identity, (d) RAF/TAP for origination, (e) Rissanen MDL. Contested/heuristic and to be cited as *implementation only*, never warrant: constructal law (Bejan), maximum-entropy-production, England's dissipation-driven adaptation, edge-of-chaos/SOC, and FEP (Friston himself calls it near-tautological). Note the shared ancestry: **Prigogine ↔ England ↔ FEP** are one non-equilibrium-thermodynamics lineage; **Ashby ↔ good-regulator ↔ internal-model ↔ FEP** are one cybernetics lineage.
- **Triality/alignment**: honest count $\approx$ **3–4 independent pillars**: (1) the differential-geometry/control definitional point that a gradient needs a datum; (2) Laplacian/Hodge consensus-and-flatness — note that **consensus-null-space and gauge-flatness are the same Hodge-theoretic fact**, so they are *one* pillar, not two; (3) the epistemic/control common-reference & common-knowledge results (Aumann; Halpern–Moses); (4) the pragmatist irreducible-triadicty philosophy (Davidson + Peirce, *partially* shared ancestry $\Rightarrow \sim 1.5$). Category theory (natural transformations / n-morphisms) is a weaker, mostly-analogical fifth; the cybernetics good-regulator chain is a contested, non-independent sixth.

Recommended headline: "a duality plus a necessary third relatum (the ground), on which alignment runs" — defensible; "irreducible triality" — only under a specified algebra.

9. Caveats

- **Analogy vs. theorem.** The rigorous results live in their own settings (Laplacian consensus, Aumann, Halpern–Moses, gauge flatness, RG, SR). Mapping them onto Γ /R/GROUND is only as strong as the claim that the architecture *instantiates* those settings.
- **The ground must be genuinely external and non-negotiable.** If RLVR is itself gameable or authored by one party, it is not a ground — it becomes another frame and the system reverts to the negotiation/power failure mode.
- **Aumann is assumption-sensitive** (common priors, the KK principle, Bayesian updating); the regress-collapse guarantee is only as strong as those hold.
- **Irreducibility is operation-relative** (Herzberger; Koshkin) — do not overclaim.
- **The dimensional limit on vantages is not established here.** Welch-type coherence bounds imply a hard ceiling on mutually distinguishable directions in a space of given dimension, which would cap useful frame count — but frames have not been shown to inhabit a well-defined metric space, so the bound is suggestive and not load-bearing. Importing it without specifying the space would be the same error the independence audit exists to catch.
- **The diversity subsidy is a live hazard.** $\lambda \cdot \text{covariance}$ rewards disagreement, and disagreement is cheap; without the ground-precedence restriction and a bounded λ it pays for being wrong in novel ways.
- **R_T is a rediscovery, and the novelty claim must be placed accordingly** (§3.6): the bracket is abstract interpretation / CEGAR / multigrid in this architecture's vocabulary. What is this work's own is the composition — MDL-scored minting as the refinement operator — not the construction.
- **Cross-scale transfer is licensed only insofar as R_T is small and structured** (§1.4). A large or unstructured R_T is not a nuisance term; it is a refutation of the analogy at that boundary, and the empirical test of whether Lane C is worth running at all.
- **Population echo independence remains unproven** (§4.3): effective independence can be *computed*, but no theorem caps achievable derivational independence under shared curriculum, bootstrap, and Γ . Report computed effective independence; do not assert independence.
- **The λ derivation carries preconditions** (§2): dwell-time guarantees stability, not

optimality; the ARL forms assume near-i.i.d. residuals; and the anti-chattering result requires the trigger statistic not be a function of λ 's own recent changes.

- **Bounded $\neq$ affordable** (§4.5): the architecture bounds branching, reachable space, search location, and acceptance, but nothing bounds whether a search *terminates with a find* within budget. That is the open empirical question.
- **Amortization can run backwards**: inheritance lowers the marginal cost only if the inherited abstractions fit the domain; where they do not, negative transfer may exceed the cost of starting from primitives.
- **"Maintaining a gradient costs energy" is false as stated** and is not claimed here: a structurally enforced difference (Donnan) is free; only non-equilibrium states require continuous expenditure. The architecture's running cost is attributed to the ground, not to frame separation.
- **Decayed coupling is not manufactured independence** (§5.1). Temporal distance and context reset remove access to correlating information while retaining its products; this is cheaper than independence and not equivalent to it, and it degrades toward source amnesia unless what is carried forward is a method rather than a conclusion.
- **Proxy substitution is the residual failure mode of an externalised ground** (§5.3), and it is not detectable from inside the system's own progress narrative — a proxy that moves supplies the same felt evidence as a metric that moves. This is the maintainer's irreducible duty and the one that grows harder as the ground grows more non-negotiable.
- **The ground-maintainer role is asserted, not derived** (§5.3). That a participating frame cannot also define the referent follows from §5.2; that no substitute exists short of a structurally non-negotiable referent is an argument, not a theorem, and it is the point at which this specification makes its largest practical demand.
- **The clean-boundary assumption is a prior, not a property of the world**. Open-ended domains produce co-occurring and nested regime changes; the architecture answers this with resolution order, probing, and quarantine (§3.5) rather than by assuming isolation. Whether probing can in practice disentangle $k > 1$ mechanisms within an action budget is untested.
- **R_T is an added mechanism, not a free signal** (§3.6): it requires deliberately round-tripping structures across scales, and its cost has not been priced against the action budget.
- **The "bill" claims and the three-primitive framing are recent** and lightly aged;

treat as [BELIEVED] until slept on and measured. ground-verified closure remains the only real metric.

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